

Readme

设计模式

一、创建型模式

1. 命令模式

- 实现了命令接口和具体命令类
- 使用 CommandFactory 创建各种命令对象
- 支持操作的撤销/重做功能

2. 单例模式

- 应用于 SessionManager 和 ConfigurationManager,使用双重检查锁定确保线程安全

```
public static synchronized SessionManager getInstance() {  
    if (instance == null) {  
        instance = new SessionManager();  
    }  
    return instance;  
}
```

- 确保系统中只有一个实例
- 提供全局访问点

3. 工厂模式

- CommandFactory 用于创建不同类型的命令对象
- 封装对象的创建过程,统一的创建接口, 隐藏具体命令类的实例化细节
- 支持多种类型的命令创建, 支持命令的扩展, 易于添加新的命令类型

4. 原型模式

- 用于 HtmlFileNode 中的文档克隆
- 支持对象的复制和克隆

5. 建造者模式

- 使用构建器用于构建文件树结构
- 实现链式调用
- 简化复杂对象的创建过程

二、结构型模式

1. 访问者模式

- DocumentVisitor 接口定义文档节点访问方法
- 实现了 TreeDisplayVisitor 和 IndentDisplayVisitor
- 支持不同的遍历显示方式

2. 组合模式

- 用于构建文件系统树结构
- FileSystemNode 作为组件接口
- HtmlFileNode 作为叶子节点
- 统一处理文件和目录

3. 代理模式

- SessionManager 作为文件系统访问的代理
- 控制对象的访问

三、行为型模式

1. 观察者模式

- 用于文件修改状态监控
- 实现状态变化的通知机制
- e.g. Editor类中执行命令时执行修改状态监控

2. 状态模式

- 管理编辑器的不同状态(Sessionstate)
- 处理状态转换

3. 备忘录模式

- 用于保存和恢复编辑器状态(SaveCommand)
- 实现状态的回退功能

4. 迭代器模式

- 用于文件系统的遍历
- 提供统一的遍历接口

5. 策略模式

- SpellChecker 接口定义拼写检查策略
- LanguageToolSpellChecker 和 DefaultSpellChecker 提供不同实现

设计原则

1. 单一职责原则

- 每个类都有明确的职责

- 如 Editor 负责文档编辑, FileTreeBuilder 负责构建文件树

2. 开闭原则

- 通过接口和抽象类实现扩展
- 支持新功能的添加而不修改现有代码
- e.g. CommandFactory支持添加新命令而不修改现有代码

3. 依赖倒置原则

- 高层模块依赖抽象接口
- 避免直接依赖具体实现
- 例如 SpellChecker 接口

4. 接口隔离原则

- 接口职责单一
- 将不同的访问者接口分开
- Command接口定义最小功能集

5. 里氏替换原则

- 子类可以替换父类
- 确保继承关系的合理性
- 例如 Visitor 、 Command 使用

6. 最少知识原则

- 减少类之间的耦合
- 通过命令工厂实现对象创建的封装
- e.g. Editor类通过Command接口与具体命令交互
- e.g. CommandFactory封装命令创建细节

7. 组合复用原则

- 优先使用组合而不是继承
- 如 Editor 包含 HtmlFileNode 而不是继承

运行方式

在demo文件夹位置执行 `java -jar demo.jar` 运行jar包, testcases文件夹中已有一些html用于测试, 执行 `load testcases/xx.html` 可以加载testcases中的html文件(xx为文件夹中已有html文件名) 当前demo文件夹中包含 `.my-html` 文件用于存储编辑器状态, 直接运行时会有已经加载的编辑器列表, load已加载的html会报错, 可以在读取的基础上继续运行, 也可以删除 `.my-html` 执行全新的编辑器

新增功能验证

load 现有文件处理

可以直接读取已有的文件

```

D:\softwareEngineering\lab2\demo\test1.html> editor-list
D:\softwareEngineering\lab2\demo\testcases\test.html
> D:\softwareEngineering\lab2\demo\test1.html
D:\softwareEngineering\lab2\demo\test1.html> load testcases/test2.html
Loaded file: testcases/test2.html
D:\softwareEngineering\lab2\demo\testcases\test2.html> dir-tree
└─ .
    ..... 省去中间的代码文件
    └─ testcases
        │   └─ innertestcases
        │   └─ newtest.html
        │   └─ test.html
        │   └─ test1.html
        └─ test2.html
    └─ demo.jar
    └─ pom.xml
    └─ readme.md
    └─ test1.html

```

load 未创建的文件新建处理

未创建的文件会直接加载初始模板

```

D:\softwareEngineering\lab2\demo\testcases\newtest.html> load test1.html
Loaded file: test1.html
D:\softwareEngineering\lab2\demo\test1.html> append h1 main-title body "Welcome to Test Page"
15:04:38.727 [main] DEBUG com.example.command.CommandHistory -- Command executed: AppendCommand
D:\softwareEngineering\lab2\demo\test1.html> print-tree
<#root>
└─ <html id="html">
    └─ <head id="head">
        └─ <title id="title">
    └─ <body id="body">
        └─ <h1 id="main-title"> Welcome to Test Page

```

load 已加载文件报错处理

针对已在编辑区的test.html，再次执行 load testcases/test.html 时出现报错

```

D:\softwareEngineering\lab2\demo\test1.html> editor-list
D:\softwareEngineering\lab2\demo\testcases\test.html
> D:\softwareEngineering\lab2\demo\test1.html *
D:\softwareEngineering\lab2\demo\test1.html> load testcases/test.html
Error: Command 'load' failed: Failed to load file: File already opened: D:\softwareEngineering\
15:45:46.581 [main] ERROR com.example.HtmlEditorApplication -- Command execution failed
com.example.exception.CommandException: Command 'load' failed: Failed to load file: File already
    at com.example.HtmlEditorApplication.handleLoad(HtmlEditorApplication.java:271)
    at com.example.HtmlEditorApplication.processCommandWithArgs(HtmlEditorApplication.java:
    at com.example.HtmlEditorApplication.processCommand(HtmlEditorApplication.java:67)
    at com.example.HtmlEditorApplication.start(HtmlEditorApplication.java:48)
    at com.example.Main.main(Main.java:5)

```

close 关闭当前活跃编辑器

编辑器成功关闭并运行剩下打开编辑器列表中的第一个

```

D:\softwareEngineering\lab2\demo\test1.html> edit testcases/newtest.html
Switched to: D:\softwareEngineering\lab2\demo\testcases\newtest.html
D:\softwareEngineering\lab2\demo\testcases\newtest.html> close
15:34:22.468 [main] DEBUG com.example.core.SessionManager -- Successfully closed editor. Open e
15:34:22.470 [main] INFO com.example.command.session.CloseCommand -- Successfully closed file: I
15:34:22.472 [main] DEBUG com.example.command.CommandHistory -- Command executed: CloseCommand
> D:\softwareEngineering\lab2\demo\testcases\test.html
D:\softwareEngineering\lab2\demo\test1.html *

```

save 保存文档

成功保存，不再出现*修改标记

```

D:\softwareEngineering\lab2\demo\test1.html> editor-list
D:\softwareEngineering\lab2\demo\testcases\test.html
> D:\softwareEngineering\lab2\demo\test1.html *
D:\softwareEngineering\lab2\demo\test1.html> save
File saved: D:\softwareEngineering\lab2\demo\test1.html
D:\softwareEngineering\lab2\demo\test1.html> editor-list
D:\softwareEngineering\lab2\demo\testcases\test.html
> D:\softwareEngineering\lab2\demo\test1.html

```

showid 控制id显示

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> print-tree
```

```
<#root>
```

```
└─ <html id="html">
  └─ <head id="head">
    └─ <title id="title"> Second Test
  └─ <body id="body">
    └─ <h1 id="title2"> Second Page
    └─ <p id="content2"> Content of second page
```

```
22:43:25.177 [main] DEBUG com.example.command.CommandHistory -- Command executed: PrintTreeComm;
```

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> showid false
```

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> print-tree
```

```
<#root>
```

```
└─ <html>
  └─ <head>
    └─ <title> Second Test
  └─ <body>
    └─ <h1> Second Page
    └─ <p> Content of second page
```

```
22:43:36.212 [main] DEBUG com.example.command.CommandHistory -- Command executed: PrintTreeComm;
```

exit 退出编辑器

退出前处理所有未保存的编辑器

```

D:\softwareEngineering\lab2\demo\testcases\test2.html> edit-id title2 changedtitle
23:07:03.924 [main] DEBUG com.example.command.CommandHistory -- Command executed: EditIdCommand
D:\softwareEngineering\lab2\demo\testcases\test2.html> editor-list
  D:\softwareEngineering\lab2\demo\testcases\test.html
> D:\softwareEngineering\lab2\demo\testcases\test2.html *
  D:\softwareEngineering\lab2\demo\test1.html
D:\softwareEngineering\lab2\demo\testcases\test2.html> edit test1.html
Switched to: D:\softwareEngineering\lab2\demo\test1.html
D:\softwareEngineering\lab2\demo\test1.html> print-tree
<#root>
├── <html>
│   ├── <head>
│   │   └── <title>
│   └── <body>
│       ├── <h1> Welcome to Test Page
│       └── <ul>
│           └── <li> First item
└──
23:07:24.981 [main] DEBUG com.example.command.CommandHistory -- Command executed: PrintTreeComm
D:\softwareEngineering\lab2\demo\test1.html> showid true
D:\softwareEngineering\lab2\demo\test1.html> print-tree
<#root>
├── <html id="html">
│   ├── <head id="head">
│   │   └── <title id="title">
│   └── <body id="body">
│       ├── <h1 id="main-title"> Welcome to Test Page
│       └── <ul id="list-1">
│           └── <li id="item1modified"> First item
└──
23:07:34.256 [main] DEBUG com.example.command.CommandHistory -- Command executed: PrintTreeComm
D:\softwareEngineering\lab2\demo\test1.html> delete list-1
23:07:40.154 [main] DEBUG com.example.command.CommandHistory -- Command executed: DeleteCommand
D:\softwareEngineering\lab2\demo\test1.html> exit
Save changes to D:\softwareEngineering\lab2\demo\testcases\test2.html? (y/n): n
Save changes to D:\softwareEngineering\lab2\demo\test1.html? (y/n): n

```

同时再次运行时可以读取之前的状态，恢复退出前的活跃编辑器及打开编辑器列表

```
D:\softwareEngineering\lab2\demo\test1.html> exit
```

```
D:\softwareEngineering\lab2\demo>java -jar demo.jar
```

HTML Editor v1.0

Type 'help' for list of commands

```
D:\softwareEngineering\lab2\demo\test1.html> showid true
```

edit 切换活跃编辑器

成功切换到另外的编辑器，并且可以在指定的编辑器执行命令

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> editor-list
```

```
D:\softwareEngineering\lab2\demo\testcases\test.html
```

```
> D:\softwareEngineering\lab2\demo\testcases\test2.html *
```

```
D:\softwareEngineering\lab2\demo\test1.html
```

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> edit test1.html
```

```
Switched to: D:\softwareEngineering\lab2\demo\test1.html
```

```
D:\softwareEngineering\lab2\demo\test1.html> print-tree
```

editor-list 显示编辑器列表

显示所有打开的编辑器，并且有**当前活跃编辑器**标记和**未保存**标记

```
D:\softwareEngineering\lab2\demo\testcases\test2.html> editor-list
```

```
D:\softwareEngineering\lab2\demo\testcases\test.html
```

```
> D:\softwareEngineering\lab2\demo\testcases\test2.html *
```

```
D:\softwareEngineering\lab2\demo\test1.html
```

print-tree 加强

在有语法错误的节点前面添加标记


```

D:\softwareEngineering\lab2\demo\testcases\test.html> print-tree
15:29:50.064 [main] DEBUG org.languagetool.AnalyzedTokenReadings -- 'MyWebpage' is immunized by
15:29:50.065 [main] DEBUG org.languagetool.AnalyzedTokenReadings -- '.' is immunized by antipati
15:29:50.065 [main] DEBUG org.languagetool.AnalyzedTokenReadings -- 'com' is immunized by antipa
<#root>
├─ <html>
│   └─ <head>
│       └─ [X] <title> Speling Test Page
│   └─ <body>
│       └─ [X] <h1> Welcom to my webpage
│       └─ [X] <p> This is a paragrph with misspeled words.
│       └─ <ul>
│           └─ [X] <li> Frist item
│           └─ [X] <li> Secund item
│           └─ [X] <li> Thrid item
│       └─ [X] <div> this is a text contnt in div
│           └─ <p> Last updated: 2024-01-01
│           └─ [X] <p> Copright ? 2021 MyWebpage.com

```

15:29:50.077 [main] DEBUG com.example.command.CommandHistory -- Command executed: PrintTreeComm

Dir-indent 目录显示

可以显示文件夹下结构并标识未保存html

```
D:\softwareEngineering\lab2\demo\test1.html> dir-indent
```

```
.
  out
    artifacts
      Html_editor_jar
      demo.jar
  src
    main
      java
        com
          example
            command
              display
                DirIndentCommand.java
                DirTreeCommand.java
                PrintIndentCommand.java
                PrintTreeCommand.java
                SpellCheckCommand.java
            edit
              AppendCommand.java
              DeleteCommand.java
              EditIdCommand.java
              EditTextCommand.java
              InsertCommand.java
            session
              CloseCommand.java
              LoadCommand.java
              SaveCommand.java
              ShowIdCommand.java
            Command.java
            CommandFactory.java
            CommandHistory.java
          config
            Configuration.java
            ConfigurationManager.java
            EditorConfig.java
            JsonConfiguration.java
          core
            Editor.java
            SessionManager.java
            SessionState.java
          display
            formatter
```

- DisplayFormatter.java
 - visitor
 - DocumentVisitor.java
 - IndentDisplayVisitor.java
 - TreeDisplayVisitor.java
 - exception
 - CommandException.java
 - DocumentException.java
 - EditorException.java
 - FileSystemException.java
 - SessionException.java
 - ValidationError.java
 - ValidationException.java
 - filesystem
 - visitor
 - FileSystemVisitor.java
 - HtmlNodeVisitor.java
 - IndentFileVisitor.java
 - TreeFileVisitor.java
 - FileSystemNode.java
 - FileTreeBuilder.java
 - HtmlFileNode.java
 - spellcheck
 - DefaultSpellChecker.java
 - LanguageToolSpellChecker.java
 - SpellChecker.java
 - SpellError.java
 - HtmlEditorApplication.java
 - Main.java
- resources
 - META-INF
 - MANIFEST.MF
- test
 - java
 - com
 - example
 - command
 - display
 - DisplayCommandsTest.java
 - edit
 - EditCommandsTest.java
 - session
 - SessionCommandsTest.java

- CommandFactoryTest.java
- CommandHistoryTest.java
- config
 - ConfigurationManagerTest.java
 - JsonConfigurationTest.java
- core
 - EditorTest.java
 - SessionManagerTest.java
 - SessionStateTest.java
- display
 - formatter
 - DisplayFormatterTest.java
 - visitor
 - IndentDisplayVisitorTest.java
 - TreeDisplayVisitorTest.java
- exception
 - CommandExceptionTest.java
 - DocumentExceptionTest.java
 - EditorExceptionTest.java
 - FileSystemExceptionTest.java
 - SessionExceptionTest.java
 - ValidationErrorTest.java
 - ValidationExceptionTest.java
- filesystem
 - visitor
 - HtmlNodeVisitorTest.java
 - IndentFileVisitorTest.java
 - TreeFileVisitorTest.java
 - FileSystemNodeTest.java
 - FileTreeBuilderTest.java
 - HtmlFileNodeTest.java
- spellcheck
 - LanguageToolSpellCheckerTest.java
 - SpellErrorTest.java
- testcases
 - innertestcases
 - newtest.html
 - test.html
 - test1.html
 - test2.html
- demo.jar
- pom.xml

```
readme.md
test1.html*
```

dir-tree 文件树显示

可以显示文件夹下结构并标识未保存html

D:\softwareEngineering\lab2\demo\test1.html> dir-tree

```
└─ .
    ├── out
    │   └── artifacts
    │       ├── Html_editor_jar
    │       └── demo.jar
    ├── src
    │   ├── main
    │   │   ├── java
    │   │   │   └── com
    │   │   │       ├── example
    │   │   │       │   ├── command
    │   │   │       │   │   ├── display
    │   │   │       │   │   │   ├── DirIndentCommand.java
    │   │   │       │   │   │   ├── DirTreeCommand.java
    │   │   │       │   │   │   ├── PrintIndentCommand.java
    │   │   │       │   │   │   ├── PrintTreeCommand.java
    │   │   │       │   │   └── SpellCheckCommand.java
    │   │   │       │   ├── edit
    │   │   │       │   │   ├── AppendCommand.java
    │   │   │       │   │   ├── DeleteCommand.java
    │   │   │       │   │   ├── EditIdCommand.java
    │   │   │       │   │   ├── EditTextCommand.java
    │   │   │       │   │   └── InsertCommand.java
    │   │   │       │   ├── session
    │   │   │       │   │   ├── CloseCommand.java
    │   │   │       │   │   ├── LoadCommand.java
    │   │   │       │   │   ├── SaveCommand.java
    │   │   │       │   │   └── ShowIdCommand.java
    │   │   │       │   ├── Command.java
    │   │   │       │   ├── CommandFactory.java
    │   │   │       │   └── CommandHistory.java
    │   │   │       └── config
    │   │   │           ├── Configuration.java
    │   │   │           ├── ConfigurationManager.java
    │   │   │           ├── EditorConfig.java
    │   │   │           └── JsonConfiguration.java
    │   │   └── core
    │   │       ├── Editor.java
    │   │       ├── SessionManager.java
    │   │       └── SessionState.java
    │   └── display
    │       └── formatter
```

```

├── DisplayFormatter.java
├── visitor
│   ├── DocumentVisitor.java
│   ├── IndentDisplayVisitor.java
│   └── TreeDisplayVisitor.java
├── exception
│   ├── CommandException.java
│   ├── DocumentException.java
│   ├── EditorException.java
│   ├── FileSystemException.java
│   ├── SessionException.java
│   ├── ValidationException.java
│   └── ValidationException.java
├── filesystem
│   ├── visitor
│   │   ├── FileSystemVisitor.java
│   │   ├── HtmlNodeVisitor.java
│   │   ├── IndentFileVisitor.java
│   │   └── TreeFileVisitor.java
│   ├── FileSystemNode.java
│   ├── FileTreeBuilder.java
│   └── HtmlFileNode.java
├── spellcheck
│   ├── DefaultSpellChecker.java
│   ├── LanguageToolSpellChecker.java
│   ├── SpellChecker.java
│   └── SpellError.java
├── HtmlEditorApplication.java
└── Main.java

resources
├── META-INF
└── MANIFEST.MF

test
├── java
│   └── com
│       └── example
│           ├── command
│           │   ├── display
│           │   │   └── DisplayCommandsTest.java
│           │   ├── edit
│           │   │   └── EditCommandsTest.java
│           └── session
│               └── SessionCommandsTest.java

```

```
|
|
|   |─ CommandFactoryTest.java
|   |─ CommandHistoryTest.java
|   └─ config
|       |─ ConfigurationManagerTest.java
|       |─ JsonConfigurationTest.java
|   └─ core
|       |─ EditorTest.java
|       |─ SessionManagerTest.java
|       |─ SessionStateTest.java
|   └─ display
|       |─ formatter
|       |   |─ DisplayFormatterTest.java
|       |─ visitor
|       |   |─ IndentDisplayVisitorTest.java
|       |   |─ TreeDisplayVisitorTest.java
|   └─ exception
|       |─ CommandExceptionTest.java
|       |─ DocumentExceptionTest.java
|       |─ EditorExceptionTest.java
|       |─ FileSystemExceptionTest.java
|       |─ SessionExceptionTest.java
|       |─ ValidationExceptionTest.java
|       |─ ValidationExceptionTest.java
|   └─ filesystem
|       |─ visitor
|       |   |─ HtmlNodeVisitorTest.java
|       |   |─ IndentFileVisitorTest.java
|       |   |─ TreeFileVisitorTest.java
|       |─ FileSystemNodeTest.java
|       |─ FileTreeBuilderTest.java
|       |─ HtmlFileNodeTest.java
|   └─ spellcheck
|       |─ LanguageToolSpellCheckerTest.java
|       |─ SpellErrorTest.java
|
|─ testcases
|   |─ innertestcases
|   |─ newtest.html
|   |─ test.html
|   |─ test1.html
|   |─ test2.html
|─ demo.jar
|─ pom.xml
```



```
|— readme.md
|— test1.html*
```